ModelIt!

Teacher's Guide

Every Drop You Drink Is Recycled Dinosaur Water

The Never-Ending Journey of Water on Earth

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-ESS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS2-4

Develop a model to describe the cycling of water through Earth's systems driven by energy from the Sun and the force of gravity.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model to describe the cycling of water through Earth's systems.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Surface Processes** — Water continually cycles among land, ocean, and atmosphere via transpiration, evaporation, condensation, crystallization, and precipitation.
Crosscutting Concept	**Energy and Matter** — Students trace the flow of energy (solar) and matter (water) through the water cycle system, showing that matter is conserved as it changes form.

CAST Testing Alignment

MULTIPLE CHOICE:

During a heat wave, evaporation rates from a lake increase significantly. Which of the following is the most likely effect on the local water cycle?

CONSTRUCTED RESPONSE:

Develop and describe a model showing how water cycles through Earth's systems. Explain the role of solar energy and gravity in driving this cycle.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water is the most important substance on Earth for supporting life. The water cycle — driven by solar energy and gravity — has been recycling the same water molecules for over 4 billion years, connecting every ocean, river, cloud, and living thing on the planet.

How the Water Cycle Works

Solar energy heats surface water in oceans, lakes, and soil, causing evaporation. Water vapor rises into the atmosphere, cools, and condenses into clouds. When droplets become heavy enough, gravity pulls them back to Earth as precipitation. Water then flows through rivers, soaks into groundwater, or is absorbed by plants — eventually returning to the ocean.

Energy Drives the Cycle

The Sun provides the energy needed to break hydrogen bonds between water molecules during evaporation. About 86% of global evaporation occurs from ocean surfaces. The energy stored in water vapor is released as latent heat when condensation occurs, driving weather patterns and storms.

Gravity's Role

Gravity is essential for precipitation — it pulls water droplets out of clouds. It also drives surface runoff (rivers flowing downhill), groundwater movement, and ocean circulation. Without gravity, water would float off into space as vapor.

Water Reservoirs

Earth's water is stored in reservoirs: oceans (97.2%), ice caps and glaciers (2.1%), groundwater (0.6%), freshwater lakes and rivers (0.01%), and atmosphere (0.001%). The cycle moves water between these reservoirs continuously. A water molecule spends an average of 3,200 years in the ocean before evaporating.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify solar energy, ground temperature, evaporation rate, and precipitation rate as key components of the water cycle system.

E — Establish Relationships
Students determine that solar energy and temperature positively drive evaporation, which positively drives precipitation.

V — Visualize & Model
Students build a model showing how energy input controls the speed and volume of water cycling.

E — Evaluate Outcomes
Students run heat wave, winter, and normal scenarios to observe how temperature changes affect cycle rates.

R — Revise & Extend
Students add wind patterns or transpiration to explore more complex water cycle dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The water you drank this morning? A dinosaur probably drank the same water 100 million years ago."*

Watch For

Let that sink in. Show a timeline of water's journey.

Slide 2: Learning Objectives

3 min

Say: *"Today we're tracking water on a journey that's been happening for 4.5 billion years."*

Watch For

Have students read objectives. Pre-teach 'transpiration.'

Slide 3: Big Question

3 min

Say: *"7.8 billion people use water every day. Agriculture uses trillions of gallons. Why don't we run out?"*

Watch For

Poll: Where does your tap water come from? Trace it backward.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the engine that recycles every drop on Earth."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What drives the water cycle? What can we control versus what responds?"*

Watch For

Guide sorting. Discuss why the Sun is the external energy source.

Slide 6: Activity 2: Connections

8 min

Say: *"More heat, more evaporation. More evaporation, more clouds. More clouds, more rain."*

Watch For

Help students trace the positive relationship chain.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens to the water cycle during a heat wave versus winter?"*

Watch For

Students predict first. Compare seasonal scenarios.

Slide 8: Discoveries

5 min

Say: *"Why is the Sun the engine of the water cycle? What role does gravity play?"*

Watch For

Discuss energy flow. Demonstrate condensation on a cold glass.

Slide 9: STEM Challenge

12 min

Say: *"You're engineers in a disaster zone. Build a system to make dirty water clean."*

Watch For

Groups build solar stills using cups, plastic wrap, and sunlight.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Solar Energy, Ground Temperature
- **INTERNAL:** Evaporation Rate, Precipitation Rate

Relationships

- **Solar Energy to Evaporation Rate:** POSITIVE (+) — More solar energy provides more heat to break hydrogen bonds in liquid water, increasing the rate of evaporation from oceans, lakes, and soil.
- **Ground Temperature to Evaporation Rate:** POSITIVE (+) — Higher ground temperatures mean warmer surface water, which evaporates more quickly. Hot pavement after rain demonstrates this effect.
- **Evaporation Rate to Precipitation Rate:** POSITIVE (+) — More evaporation puts more water vapor into the atmosphere, which eventually condenses and falls as increased precipitation.

Key Discoveries

- The Sun is the engine of the water cycle — without solar energy, water wouldn't evaporate
- Gravity pulls precipitation back to Earth and drives groundwater flow downhill
- The same water molecules have been cycling through Earth's systems for billions of years
- Temperature controls the SPEED of the cycle — hotter conditions accelerate it

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"New water is constantly being created"	Earth has had roughly the same amount of water for 4 billion years. No new water is being created — it's being RECYCLED endlessly. The water cycle moves water between reservoirs but doesn't create or destroy it.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Clouds are made of water vapor"	Clouds are made of tiny LIQUID water droplets or ice crystals — not water vapor. Water vapor is invisible gas. When we see a cloud, the vapor has already CONDENSED into visible droplets.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Rain comes from the ocean directly above"	Rain can fall thousands of miles from where the water originally evaporated. Wind carries water vapor across continents before it condenses and falls. Water that evaporated from the Pacific Ocean can fall as rain in Kansas.